02-20000-17

Original Effective Date: 09/01/01

Reviewed: 08/27/26

Revised: 09/15/26

Subject: Autologous Chondrocyte Implantation (ACI)

THIS MEDICAL COVERAGE GUIDELINE IS NOT AN AUTHORIZATION, CERTIFICATION, EXPLANATION OF BENEFITS, OR A GUARANTEE OF PAYMENT, NOR DOES IT SUBSTITUTE FOR OR CONSTITUTE MEDICAL ADVICE. ALL MEDICAL DECISIONS ARE SOLELY THE RESPONSIBILITY OF THE PATIENT AND PHYSICIAN. BENEFITS ARE DETERMINED BY THE GROUP CONTRACT, MEMBER BENEFIT BOOKLET, AND/OR INDIVIDUAL SUBSCRIBER CERTIFICATE IN EFFECT AT THE TIME SERVICES WERE RENDERED. THIS MEDICAL COVERAGE GUIDELINE APPLIES TO ALL LINES OF BUSINESS UNLESS OTHERWISE NOTED IN THE PROGRAM EXCEPTIONS SECTION.

Position Statement	Billing/Coding	Reimbursement	Program Exceptions	Definitions	Related Guidelines
Other	References	Updates			

DESCRIPTION:

Damaged articular cartilage typically fails to heal on its own and can be associated with pain, loss of function, and disability and may lead to debilitating osteoarthritis over time. These manifestations can severely impair activities of daily living and adversely affect quality of life.

With autologous chondrocyte implantation (ACI), a region of healthy articular cartilage is identified and biopsied through arthroscopy. The tissue is sent to a facility licensed by the U.S. Food and Drug Administration (FDA) where it is minced and enzymatically digested, and the chondrocytes are separated by filtration. The isolated chondrocytes are cultured for 11 to 21 days to expand the cell population, tested, and then shipped back for implantation. Under general anesthesia, an arthrotomy is performed, and the chondral lesion is excised up to the normal surrounding cartilage. Methods to improve the first-generation ACI procedure have been developed, including the use of a scaffold or matrix-induced autologous chondrocyte implantation (MACI), composed of biocompatible carbohydrates, protein polymers, or synthetics. The only FDA-approved MACI product to date is supplied in a sheet, which is cut to size and fixed with fibrin glue. This procedure is considered technically easier and less time-consuming than the first-generation technique, which required suturing of a periosteal or collagen patch and injection of chondrocytes under the patch.

A number of second-generation methods for implanting autologous chondrocytes in a biodegradable matrix are currently in development or testing or are available outside of the United States. In 2020, the FDA granted breakthrough status to Agili-C® (CartiHeal, Ltd.), a proprietary cell-free biocompatible and biodegradable tapered-shape implant for the treatment of cartilage lesions in arthritic and non-arthritic joints that, when implanted into a pre-prepared osteochondral hole, acts as a 3- dimensional scaffold that potentially supports and promotes the regeneration of the articular cartilage and its underlying subchondral bone. Agili-C was FDA-approved in 2021 for the treatment of knee-joint surface lesions with a treatable area of 1 to 7 cm2 without severe osteoarthritis.

Summary and Analysis of Evidence: Bartlett et al (2005) reported on a randomized comparison between MACI and autologous chondrocyte implantation with a collagen cover in 91 patients. Overall, results were comparable for both treatments. The modified Cincinnati Knee Rating System score improved by 17.6 points in the autologous chondrocyte implantation group and by 19.6 points in the MACI group. Visual analog scale scores improved from 6.0 to 4.3 in the autologous chondrocyte implantation group and from 6.0 to 4.1 in the MACI group. Factors associated with worse clinical outcomes were a failed prior procedure, duration of symptoms, and patient age. Second-look arthroscopy at 1 year for 42

patients showed excellent-to-good International Cartilage Repair Society scores in 79.2% of autologous chondrocyte implantation and in 66.6% of MACI patients. The authors did not report whether the study was adequately powered for this comparison. Histology from 14 autologous chondrocyte implantation and 11 MACI patients showed similar percentages of hyaline-like cartilage (42.9% autologous chondrocyte implantation, 36.4% MACI).

The SUMMIT trial was the pivotal, industry-sponsored, multicenter randomized open-label trial; it was reported by Saris et al (2014) and compared MACI with microfracture (MF) for larger cartilage defects (≥3 cm2), which typically fare worse than smaller lesions when treated with MF. 144 patients included had at least 1 symptomatic grade III or IV focal cartilage defect on the femoral condyles or trochlea, a stable knee, an intact or partial meniscus, and a moderate-to-severe KOOS pain value (<55). The average lesion size was 4.8 cm2 (range, 3 to 20 cm2), and 34.6% of patients had undergone a prior marrow stimulation procedure. At 2-year follow-up, the MACI group had significantly better subscores for KOOS pain and function in sport and recreation, as well as the other KOOS subscales (function in daily living, knee-related quality of life, other symptoms). With response to treatment defined as a 10-point improvement in both the KOOS pain and function subscales, significantly more patients in the MACI group responded to treatment (87.5%) than in the MF group (68.1%). There were no significant differences between groups for cartilage repair, as measured by second-look arthroscopy, biopsy, or magnetic resonance imaging (MRI).

A 2022 systematic review (Angele et al, 2022) reported outcomes of randomized trials of cartilage repair techniques for localized cartilage defects of the knee with a minimum 5-year follow-up. The 6 included RCTs comprised 520 patients, with mean follow-up ranging from 5 to 16 years. One trial (SUMMIT) compared MACI to MF, and 3 compared other autologous chondrocyte implantation techniques to either MF or osteochondral autograft transplantation. The trial comparing MACI to MF indicated superior outcomes in the Knee Injury and Osteoarthritis Outcome Score (KOOS) pain, function, and activities of daily living subscales with MACI; trials of other autologous chondrocyte implantation modalities produced mixed results, with 2 trials indicating no difference relative to MF in overall KOOS or other patient-reported outcome measures. One trial indicated significant improvement in overall KOOS relative to MF in a subgroup of patients with symptom onset within 3 years prior to intervention, and 1 trial indicated superior Cincinatti Knee Rating System scores at 10-year follow-up relative to osteochondral autograft transfer.

Abraamyan et al (2022) completed a systematic review with meta-analysis that evaluated cartilage repair techniques, including MF, augmented MF, and autologous chondrocyte implantation/MACI. The authors included a total of 14 RCTs (N=775), and changes from baseline in the 5 KOOS subscales, including KOOS Sport, KOOS Quality of Life, KOOS Symptoms, KOOS Pain, and KOOS Activities of Daily Living, were measured. Only the KOOS Sport subscale demonstrated statistically significant benefits with autologous chondrocyte implantation/MACI procedures compared with MF. The mean delta KOOS Sport after autologous chondrocyte implantation/MACI procedures was 9.9 points greater than after MF and 11.7 points greater than after augmented MF. Comparisons between surgical techniques for the other subscales did not reach statistical significance.

For large lesions, autologous chondrocyte implantation results in better outcomes than MF, particularly in the long term. Studies comparing autologous chondrocyte implantation with osteochondral autograft transfer have shown similar outcomes with smaller lesions, and improved outcomes with autologous chondrocyte implantation when a defect is greater than 4 cm2.

A 2022 systematic review with Bayesian network meta-analysis by Migliorini, Maffulli et al, 2022 evaluated 13 studies with a minimum 18-month follow-up comparing surgical interventions for chondral defects of the talus. The studies comprised 521 patients, with a median follow-up of 47.8 months; most studies, including all that evaluated autologous chondrocyte implantation, were retrospective, with 1 RCT and 2 prospective cohort trials included. The authors found that cell-free autologous membrane-induced chondrogenesis produced the highest American Orthopedic Foot and Ankle Society (AOFAS) scores and produced the lowest rates of failure. However, the timeframe for reporting of AOFAS score and other endpoints was not described, and funnel plots for all reported outcomes suggest the presence of publication bias.

Hu et al (2021) reported a systematic review with meta-analysis of studies published through November 2020. The authors included a total of 23 case series (N=458) with a mean duration of 12 to 154.8 months. In 6 studies, periosteum-covered autologous chondrocyte implantation was applied while 17 studies used second-generation MACI. Results demonstrated an 89% success rate (AOFAS score >80) with autologous chondrocyte implantation. Furthermore, AOFAS scores significantly improved after treatment. Krueger et al (2023) reported a retrospective case series of 36 consecutive patients who underwent autologous chondrocyte implantation for cartilage defects of the acetabulum. With a mean follow-up of 29.9 months (minimum 24 months), the mean modified Harris Hip Score improved significantly between the preoperative baseline and last follow-up, and the mean patient-reported Subjective Hip Value improved from 51.5% at pre-operative baseline to 87.4% postoperatively. The authors stated no serious intraoperative complications or postoperative adverse events were observed.The evidence on the use of autologous chondrocyte implantation for joints other than the knee includes case series, systematic reviews of case series, and a network meta-analysis of prospective and retrospective studies (no prospective studies evaluated autologous chondrocyte implantation). The most commonly reported use of autologous chondrocyte implantation is for the talus; one case series describes use for the acetabulum. Comparative trials are needed to determine whether autologous chondrocyte implantation improves outcomes for lesions of the talus and other joints.

Current research in pig-to-human xenotransplantation includes strategies to overcome the barrier of hyperacute rejection by preventing or inhibiting complement activation caused by Gal- and non-Gal-epitopes. Promising approaches to modulate hyperacute rejection involve gene knockout of GGTA1 and Neu5Gc gene and transgenic expression of hCregs. Chondrocytes from multi-genetically modified pigs might represent a promising cell source for cartilage repair because they are protected from humoral rejection in vitro. While cell viability could be completely preserved with exception of single GalTKO, the residual amount of non-lytic complement activation must be addressed by additional strategies (Tritschler et al, 2021).

Altschuler et al (2023) examined the potential superiority of a novel 3-dimensionally structured off-the-shelf scaffold (Agili-C; Cartiheal Ltd) over the current surgical standard of care (SSOC) (arthroscopic debridement or microfracture) in the treatment of knee chondral and osteochondral lesions. A total of 251 patients were enrolled in 26 medical centers according to the following criteria: age 21 to 75 years, up to 3 cartilage defects of International Cartilage Regeneration & Joint Preservation Society grade 3a or above located on the femoral condyles and/or trochlea, total treatable area from 1 to 7 cm2, bony defect depth ≤8 mm, and knee osteoarthritis grade 0 to 3 according to Kellgren-Lawrence score. Patients were randomized to the aragonite-based implant or debridement/microfracture control arm in a 2:1 ratio. Evaluation was performed at 6, 12, 18, and 24 months based on overall Knee injury and Osteoarthritis Outcome Score (KOOS) as the primary endpoint, and the KOOS subscales (Pain, Quality of Life, Activities of Daily Living), percentage of responders, and International Knee Documentation Committee (IKDC) subjective score as the secondary endpoints. Patients also underwent magnetic resonance imaging evaluation at 12 and 24 months to assess defect fill grade. Failures (ie, need for any secondary treatment) and adverse events were also recorded. The implant group showed a statistically superior outcome in the primary endpoint and all secondary endpoints at each follow-up. The magnitude of improvement in the implant group was twice as large as that in the control group in terms of mean KOOS improvement at 2 years. Responder rate (defined as at least a 30-point improvement in overall KOOS) was 77.8% in the implant group as opposed to 33.6% in the control. Statistically superior results were seen in the IKDC score as well. At 24 months, 88.5% of the implanted group had at least 75% defect fill on magnetic resonance imaging as compared with 30.9% of controls. The failure rate was 7.2% for the implant group versus 21.4% for control. The authors concluded the “aragonite-based scaffold was safe and effective in the treatment of chondral and osteochondral lesions in the knee, including patients with mild to moderate osteoarthritis, and provided superior outcomes as compared with the control group.” The authors noted study design limitations that included unbalanced parameters related to groups being very similar in baseline characteristics; prevalence of mild to moderate osteoarthritis in the SSOC group was slightly higher than in the implant arm; and the implant arm included more patients affected by deep osteochondral defects and more patients with larger lesions. They also noted the choice of debridement/microfracture as the control group, as suggested by FDA, leaves space for additional comparative studies against other approaches to further confirm the superiority of this scaffold. A 2022 ECRI Clinical Evidence Assessment, “Agili-C Cell-free Osteochondral Scaffold (Smith & Nephew, Inc.) for

Knee Cartilage Repair" stated "Agili-C is intended to repair chondral and osteochondral defects in joints by stimulating cartilage growth. Agili-C is intended as an alternative to standard surgical treatments (e.g., debridement, microfracture), autografts, biologic and cell-based regenerative treatments, and use of other natural, tissue-engineered, or inorganic grafts." The review indicated that evidence from one randomized control trial (RCT) and four pre-post studies indicates Agili-C improves knee function and activities of daily living. The RCT also indicates that Agili-C improves patient-oriented outcomes more than surgical treatment (e.g., debridement, microfracture). The review noted evidence limitations, stating "(o)nly one study compared Agili-C with SSOC, and one other study compared Agili-C with other implants. No studies reported longer-term outcomes (i.e., >5 years), which is needed to determine the treatment's long-term safety and efficacy. All studies had small sample sizes and focused on a narrow demographic patient population. The studies were not blinded; therefore, self-reported patient-oriented outcomes in the RCT comparing Agili-C and SSOC are at risk of bias. Larger RCTs that include wider and more balanced patient demographic (i.e., age, gender, activity level) that compare Agili-C with SSOC and competing implants would help support stronger and more generalizable conclusions."

POSITION STATEMENT:

Autologous chondrocyte implantation (ACI) with autologous cultured chondrocytes (e.g., Carticel), **OR** matrix-induced chondrocyte implantation (MACI) autologous cultured chondrocytes on porcine collagen membrane (e.g., Vericel) **meets the definition of medical necessity** when **ALL** of the following are met:

- Disabling full thickness articular cartilage defects of the knee caused by acute or repetitive trauma
- Adolescents should be skeletally mature with documented closure of growth plates (eg, ≥15 years)
- Adults should be too young to be considered an appropriate candidate for total knee arthroplasty or other reconstructive knee surgery (eg, < 55 years)
- Focal, full-thickness (grade III or IV) unipolar lesions of the weight-bearing surface of the femoral condyles, trochlea, or patella at least 1.5 cm2 in size
- Documented minimal to absent degenerative changes in the surrounding articular cartilage (Outerbridge grade II or less), and normal-appearing hyaline cartilage surrounding the border of the defect
- Normal knee biomechanics or alignment and stability achieved concurrently with autologous chondrocyte implantation

Autologous chondrocyte implantation (ACI) for all other joints, including but not limited to the shoulder, hip, tibia, ankle, and talus, is considered **experimental or investigational**. There is a lack of clinical scientific evidence published in peer-reviewed literature to permit conclusions on safety and net health outcomes.

Autologous chondrocyte implantation (ACI) with synthetic resorbable polymers/scaffolds (eg, AGILI-C®) is considered **experimental or investigational**. Data in published medical literature are inadequate to permit scientific conclusions on long-term and net health outcomes.

*****Outerbridge Arthroscopic Grading System**

Grade 0: Normal cartilage

Grade I: Softening and swelling/blistering

Grade II: Partial thickness defect, fissures < 1.5cm diameter/wide

Grade III: Fissures /defects down to subchondral bone with intact calcified cartilage layer, diameter > 1.5cm

Grade IV: Exposed subchondral bone

BILLING/CODING INFORMATION:

CPT Coding:

27412	Autologous chondrocyte implantation, knee
0737T	Xenograft implantation into the articular surface **(Investigational)**

HCPCS Coding:

J7330	Autologous cultured chondrocytes, implant
S2112	Arthroscopy, knee, surgical for harvesting of cartilage (chondrocyte cells)

REIMBURSEMENT INFORMATION:

Refer to section entitled **POSITION STATEMENT**.

PROGRAM EXCEPTIONS:

Federal Employee Program (FEP): Follow FEP guidelines.

State Account Organization (SAO): Follow SAO guidelines.

Medicare Advantage products: No National Coverage Determination (NCD) and/or Local Coverage Determination (LCD) were found at the time of the last guideline reviewed date.

If this Medical Coverage Guideline contains a step therapy requirement, in compliance with Florida law 627.42393, members or providers may request a step therapy protocol exemption to this requirement if based on medical necessity. The process for requesting a protocol exemption can be found at Coverage Protocol Exemption Request.

DEFINITIONS:

Autologous: derived from the implantation recipient.

Chondrocyte: mature cells found in cartilage.

RELATED GUIDELINES:

Unicondylar Interpositional Spacer Devices, 02-2000-26

OTHER:

None applicable.

REFERENCES:

1. Abraamyan T, Johnson AJ, Wiedrick J, Crawford DC. Marrow Stimulation Has Relatively Inferior Patient-Reported Outcomes in Cartilage Restoration Surgery of the Knee: A Systematic Review and Meta-analysis of Randomized Controlled Trials. Am J Sports Med. 2022 Mar;50(3):858-866. doi: 10.1177/03635465211003595. Epub 2021 Apr 23. PMID: 33890799.
2. Altschuler N, Zaslav KR, Di Matteo B, Sherman SL, Gomoll AH, Hacker SA, Verdonk P, Dulic O, Patrascu JM, Levy AS, Robinson D, Kon E. Aragonite-Based Scaffold Versus Microfracture and Debridement for the Treatment of Knee Chondral and Osteochondral Lesions: Results of a

Multicenter Randomized Controlled Trial. Am J Sports Med. 2023 Mar;51(4):957-967. doi: 10.1177/03635465231151252. Epub 2023 Feb 13.

3. American Academy of Orthopaedic Surgeons. Clinical practice guideline on the diagnosis and treatment of osteochondritis dissecans. 2010. Accessed at http://www.aaos.org/research/guidelines/OCD_guideline.pdf.
4. Angele P, Zellner J, Schröter S, Flechtenmacher J, Fritz J, Niemeyer P. Biological Reconstruction of Localized Full-Thickness Cartilage Defects of the Knee: A Systematic Review of Level 1 Studies with a Minimum Follow-Up of 5 Years. Cartilage. 2022 Dec;13(4):5-18. doi: 10.1177/19476035221129571. Epub 2022 Oct 15.
5. Bartlett W, Skinner JA, Gooding CR, Carrington RW, Flanagan AM, Briggs TW, Bentley G. Autologous chondrocyte implantation versus matrix-induced autologous chondrocyte implantation for osteochondral defects of the knee: a prospective, randomised study. J Bone Joint Surg Br. 2005 May;87(5):640-5. doi: 10.1302/0301-620X.87B5.15905.
6. Basad E, et al. Matrix-induced autologous chondrocyte implantation versus microfracture in the treatment of cartilage defects of the knee: a 2-year randomised study. Knee surgery, sports traumatology, arthroscopy: official journal of the ESSKA 18.4 (2010): 519.
7. Baums MH, Schultz W, Kostuj T, Klinger HM. Cartilage repair techniques of the talus: An update. World J Orthop. 2014 Jul 18;5(3):171-9.
8. Blue Cross and Blue Shield Association Technology Evaluation Center (TEC) – Autologous Chondrocyte Transplantation TEC Assessments 2003, Tab 2.
9. Blue Cross Blue Shield Association. 7.01.48-Autologous Chondrocyte Implantation for Focal Articular Cartilage Lesions, 05/26.
10. Buckwalter JA, Bowman GN, Albright JP, Wolf BR, Bollier M. Clinical outcomes of patellar chondral lesions treated with juvenile particulated cartilage allografts. Iowa Orthop J. 2014;34:44-9.
11. Clar C, Cummins E, McIntyre L, Thomas S, Lamb J, Bain L, Jobanputra P, Waugh N. Clinical and cost-effectiveness of autologous chondrocyte implantation for cartilage defects in knee joints: systematic review and economic evaluation. Health Technology Assessment Vol.9: No.47. 2005:98. The National Coordinating Centre for Health Technology Assessment (NCCHTA).
12. Conte P, Anzillotti G, Crawford DC, Dasa V, Flanigan DC, Nordt WE, Scopp JM, Meislin RJ, Strauss EJ, Strickland SM, Fiorentino G, Lattermann C. Differential analysis of the impact of lesions' location on clinical and radiological outcomes after the implantation of a novel aragonite-based scaffold to treat knee cartilage defects. Int Orthop. 2024 Dec;48(12):3117-3126. doi: 10.1007/s00264-024-06314-1. Epub 2024 Sep 21.
13. Cuéllar R, Cuéllar A, Ponte J, Ruiz-Ibán MA. Arthroscopic technique for the treatment of patellar chondral lesions with the patient in the supine position. Arthrosc Tech. 2014 Jun 2;3(3):e373-6.
14. Davies RL, Kuiper NJ. Regenerative Medicine: A Review of the Evolution of Autologous Chondrocyte Implantation (ACI) Therapy. Bioengineering (Basel). 2019;6(1):22. Published 2019 Mar 13. doi:10.3390/bioengineering6010022.
15. Dunkin BS, Lattermann C. New and Emerging Techniques in Cartilage Repair: MACI. Oper Tech Sports Med. 2013 Jun 1;21(2):100-107.
16. ECRI Clinical Evidence Assessment. Agili-C Cell-free Osteochondral Scaffold (Smith & Nephew, Inc.) for Knee Cartilage Repair (June 2022).
17. ECRI Health Technology Assessment Information Services – Health Technology Assessment News – "Long Term Results of Autologous Chondrocyte Transplantation" (May-June 2000).
18. ECRI Health Technology Assessment Information Services – Windows on Medical Technology – "Autologous Chondrocyte Implantation for Knee Cartilage Defects" (July 2004. Issue No. 112).

19. Gikas PD, Morris T, Carrington R, Skinner J, Bentley G, Briggs T. A correlation between the timing of biopsy after autologous chondrocyte implantation and the histological appearance. J Bone Joint Surg Br. 2009 Sep; 91(9):1172-7.

20. Gobbi A, et al. Matrix-Induced Autologous Chondrocyte Implantation versus Multipotent Stem Cells for the Treatment of Large Patellofemoral Chondral Lesions: A Nonrandomized Prospective Trial. Cartilage. 2015 Apr;6(2):82-97.

21. Harris JD, et al. Failures, re-operations, and complications after autologous chondrocyte implantation – a systematic review. Osteoarthritis and cartilage 19.7 (2011): 779-791.

22. Hayes, Inc Hayes Medical Technology Directory – "Autologous Chondrocyte Transplantation of the Knee" Lansdale, PA: Hayes, Inc.; January 2005. Update performed 02/01/07.

23. Hoburg A, Löer I, Körsmeier K, et al. Matrix-Associated Autologous Chondrocyte Implantation Is an Effective Treatment at Midterm Follow-up in Adolescents and Young Adults. Orthop J Sports Med. 2019;7(4):2325967119841077. Published 2019 Apr 25. doi:10.1177/2325967119841077.

24. Hu M, Li X, Xu X. Efficacy and safety of autologous chondrocyte implantation for osteochondral defects of the talus: a systematic review and meta-analysis. Arch Orthop Trauma Surg. 2023 Jan;143(1):71-79. doi: 10.1007/s00402-021-03990-1. Epub 2021 Jun 14. PMID: 34128117.

25. Kraeutler MJ, Belk JW, Purcell JM, McCarty EC. Microfracture Versus Autologous Chondrocyte Implantation for Articular Cartilage Lesions in the Knee: A Systematic Review of 5-Year Outcomes. Am J Sports Med. 2018;46(4):995-999. doi:10.1177/0363546517701912. PMID: 28423287.

26. Krueger DR, Baur ADJ, Perka C, Schroeder JH. Injectable autologous chondrocyte implantation in acetabular cartilage defects: 2-year minimum clinical and MRI results. Arch Orthop Trauma Surg. 2023 Feb;143(2):739-747. doi: 10.1007/s00402-021-04141-2. Epub 2021 Sep 1. PMID: 34468836.

27. Lee YH, Suzer F, Thermann H. Autologous Matrix-Induced Chondrogenesis in the Knee: A Review. Cartilage. 2014 Jul;5(3):145-53.

28. Macmull S, Jaiswal PK, Bentley G, Skinner JA, Carrington RW, Briggs TW. The role of autologous chondrocyte implantation in the treatment of symptomatic chondromalacia patellae. Int Orthop. 2012 Jul;36(7):1371-7.

29. Mancò A, et al. Microfracture versus microfracture and platelet-rich plasma: arthroscopic treatment of knee chondral lesions. A two-year follow-up study. Joints 4.3 (2016): 142.

30. Marlovits S, Aldrian S. Clinical and Radiological Outcomes 5 Years After Matrix-Induced Autologous Chondrocyte Implantation in Patients With Symptomatic, Traumatic Chondral Defects. Am J Sports Med. 2012 Oct;40(10):2273-80.

31. Merkely G, Ogura T, Ackermann J, Barbieri Mestriner A, Gomoll AH. Clinical Outcomes after Revision of Autologous Chondrocyte Implantation to Osteochondral Allograft Transplantation for Large Chondral Defects: A Comparative Matched-Group Analysis. Cartilage. 2021 Apr;12(2):155-161. doi: 10.1177/1947603519833136. Epub 2019 Mar 22.

32. Migliorini F, Maffulli N, Schenker H, Eschweiler J, Driessen A, Knobe M, Tingart M, Baroncini A. Surgical Management of Focal Chondral Defects of the Talus: A Bayesian Network Meta-analysis. Am J Sports Med. 2022 Aug;50(10):2853-2859. doi: 10.1177/03635465211029642. Epub 2021 Sep 20.

33. Mistry H, Connock M, Pink J, et al. Autologous chondrocyte implantation in the knee: systematic review and economic evaluation. Health Technol Assess. 2017;21(6):1-294. doi:10.3310/hta21060.

34. Nassar JE, Guerin G, Keel T, Russo R, Familiari F, Tollefson LV, LaPrade RF. Autologous chondrocyte implantation, matrix-induced autologous chondrocyte implantation, osteochondral autograft transplantation and osteochondral allograft improve knee function and pain with considerations for patient and cartilage defects characteristics: A systematic review and meta-analysis. Knee Surg Sports Traumatol Arthrosc. 2025 Aug;33(8):2745-2762. doi: 10.1002/ksa.12525. Epub 2024 Nov 4.

35. National Imaging Associates, Inc. Knee Arthroscopy; Open, Non-Arthroplasty Knee Repair & Manipulation Procedures Clinical Guideline, 2018.

36. National Institute for Health and Care Excellence (NICE). Interventional procedures guidance IPG792: Single-step scaffold insertion for repairing symptomatic chondral knee defects (September 2024; updated August 2025). Accessed at https://www.nice.org.uk/.

37. National Institute for Clinical Excellence. The use of autologous chondrocyte implantation for the treatment of cartilage defects in knee joints. London: National Institute for Clinical Excellence (NICE). 2005:28. National Institute for Clinical Excellence (NICE).

38. Oehme S, Burger JA, Krafzick S, Bartek B, Winkler T, Jung T. Autologous bone grafting in combination with autologous chondrocyte implantation yields favourable outcomes in the treatment of osteochondral defects of the knee: A systematic literature review. Knee Surg Sports Traumatol Arthrosc. 2024 Nov;32(11):2999-3014. doi: 10.1002/ksa.12342. Epub 2024 Jul 2.

39. Perera JR, Gikas PD, Bentley G. The present state of treatments for articular cartilage defects in the knee. Ann R Coll Surg Engl. 2012 Sep;94(6):381-7.

40. Saris D, Price A, Widuchowski W, Bertrand-Marchand M, Caron J, Drogset JO, Emans P, Podskubka A, Tsuchida A, Kili S, Levine D, Brittberg M; SUMMIT study group. Matrix-Applied Characterized Autologous Cultured Chondrocytes Versus Microfracture: Two-Year Follow-up of a Prospective Randomized Trial. Am J Sports Med. 2014 Jun;42(6):1384-94. doi: 10.1177/0363546514528093. Epub 2014 Apr 8. PMID: 24714783.

41. Seo SS, Kim CW, Jung DW. Management of focal chondral lesion in the knee joint. Knee Surg Relat Res. 2011 Dec;23(4):185-96.

42. Teo BJ, Buhary K, Tai BC, Hui JH. Cell-based therapy improves function in adolescents and young adults with patellar osteochondritis dissecans. Clin Orthop Relat Res. 2013 Apr;471(4):1152-8.

43. Tritschler H, Fischer K, Seissler J, Fiedler J, Halbgebauer R, Huber-Lang M, Schnieke A, Brenner RE. New Insights into Xenotransplantation for Cartilage Repair: Porcine Multi-Genetically Modified Chondrocytes as a Promising Cell Source. Cells. 2021 Aug 20;10(8):2152. doi: 10.3390/cells10082152.

44. UpToDate. Management of osteochondritis dissecans (OCD). 2025. Accessed at uptodate.com.

45. UpToDate. Overview of surgical therapy of knee and hip osteoarthritis. 2025. Accessed at uptodate.com.

46. Vannini F, Cavallo M, Baldassarri M, Castagnini F, Olivieri A, Ferranti E, Buda R, Giannini S. Treatment of juvenile osteochondritis dissecans of the talus: current concepts review. Joints. 2015 Feb 13;2(4):188-91.

47. Vonk LA, Roël G, Hernigou J, Kaps C, Hernigou P. Role of Matrix-Associated Autologous Chondrocyte Implantation with Spheroids in the Treatment of Large Chondral Defects in the Knee: A Systematic Review. Int J Mol Sci. 2021 Jul 1;22(13):7149. doi: 10.3390/ijms22137149.

48. Washington State Health Care Authority. Health Technology Assessment Program (HTA). Treatment of Chondral Defects of the Knee: Final Evidence Report (August 21, 2024).

49. Wasiak J, Clar C, Villanueva E. Autologous cartilage implantation for full thickness articular cartilage defects of the knee. Cochrane Database of Systematic Reviews 2006, Issue 3. Art. No.: CD003323. DOI: 10.1002/14651858.CD003323.pub2.

50. Wen J, Syed B, Ansari U, et al. Favorable Short-Term Outcomes of Matrix-Associated Autologous Chondrocyte Implantation for Osteochondral Lesions of the Talus: A Systematic Review. Arthroscopy. 2025 Dec;41(12):5492-5502.e1. [Abstract]

51. Zhang Z, et al.Matrix-induced autologous chondrocyte implantation for the treatment of chondral defects of the knees in Chinese patients. Drug Des Devel Ther. 2014 Dec 5;8:2439-48.

COMMITTEE APPROVAL:

This Medical Coverage Guideline (MCG) was approved by the Florida Blue Medical Policy and Coverage Committee on 08/27/26.

GUIDELINE UPDATE INFORMATION:

09/01/01	Coding changes.
01/01/02	Coding changes.
02/15/02	Various revisions.
03/15/03	Reviewed; revised to remove unrelated coding.
08/15/03	Coverage statement for ACI changed; meniscal allograft information removed and separate MCG developed; unrelated coding information removed.
01/01/04	HCPCS coding update.
07/15/04	Review and revision of guideline; consisting of updated references.
01/01/05	Annual HCPCS update; consisting of the addition of 27412 and deletion of S2113.
07/15/05	Review and revision of guideline; consisting of updates references.
09/15/07	Review and revision of guideline consisting of updated references and reformatted guideline.
10/15/09	Scheduled review; no change in position statement.
05/11/14	Revision: Program Exceptions section updated.
03/15/15	Scheduled review. Revised description, position statement, CPT/HCPCS coding, and definitions. Updated references and reformatted guideline.
12/15/15	Revision; updated description section; added coverage for lesions of the patella, and deleted requirement for prior surgical procedure. Added coverage statement (E/I) for juvenile cartilage allografts. Updated index terms and references. Reformatted guideline.
08/15/17	Revision: updated description section. Deleted E/I coverage statement for matrix-induced autologous chondrocyte implantation (MACI). Updated references. Reformatted guideline.
07/15/18	Scheduled review. Revised description section. Revised criteria for autologous chondrocyte implantation. Added Kellgren-Lawrence Grading System. Updated references.
07/15/20	Scheduled review. Revised position statement and updated references.
03/15/22	Scheduled review. Revised description and HCPCS coding. Maintained position statement and updated references.
07/01/22	Quarterly CPT/HCPCS coding update. Added 0737T.
05/23/23	Update to Program Exceptions section.
01/01/24	Position statements maintained.
09/15/24	Scheduled review. Revised description, maintained position statement and updated references.
09/15/25	Scheduled review. Revised description, maintained position statement and updated references.
12/15/25	Revision. Revised description, added coverage statement for synthetic resorbable polymers/scaffolds (eg, AGILI-C®), and updated references.
09/15/26	Review; no change in position statement. Updated references.